An expert system for detecting automobile insurance fraud using social network analysis

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INTRODUCTION TO THE PAPER

AUTHORS

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AIM OF THE PAPER

To build a model that can detect insurance automobile fraud, focusing on finding single entities inside groups of fraudsters.

SCENARIO

In the automobile insurance sector, around 20% of all claims are fraudulent.

NETWORK REPRESENTATION

Lack of large datasets and importance of visualization.

A BRIEF VIEW ON RELATED WORKS

MACHINE LEARNING ALGORITHMS

For example: Neural Networks, Support Vector Machine, Random Forest, etc...

Cons: wrong interpretation of data; lack of labeled-unlabeled data.

SOCIAL NETWORKS

For example the work made by Noble et Al (2003): Graph-based anomaly detection.

Cons: it just considers structures; works better on larger networks.

NETWORKS AND CLASSIFICATION

For example: Relaxation Labeling. Classification of an entity related to other entities (inference).

Cons: also requiring learning from a large labeled dataset.

SOME THEORY OF SOCIAL NETWORKS

WHAT IS A NETWORK

MULTIGRAPH

LABELED GRAPH

CONNECTED COMPONENTS

DEGREE OF A NODE

PATH & GEODESICS

CENTRALITY

(betweenness centrality, closeness centrality, eigen centrality

DISTANCE

DIAMETER

VERTEX COVER

COMMUNITIES

(detected by edge betweenness)

THE SYSTEM

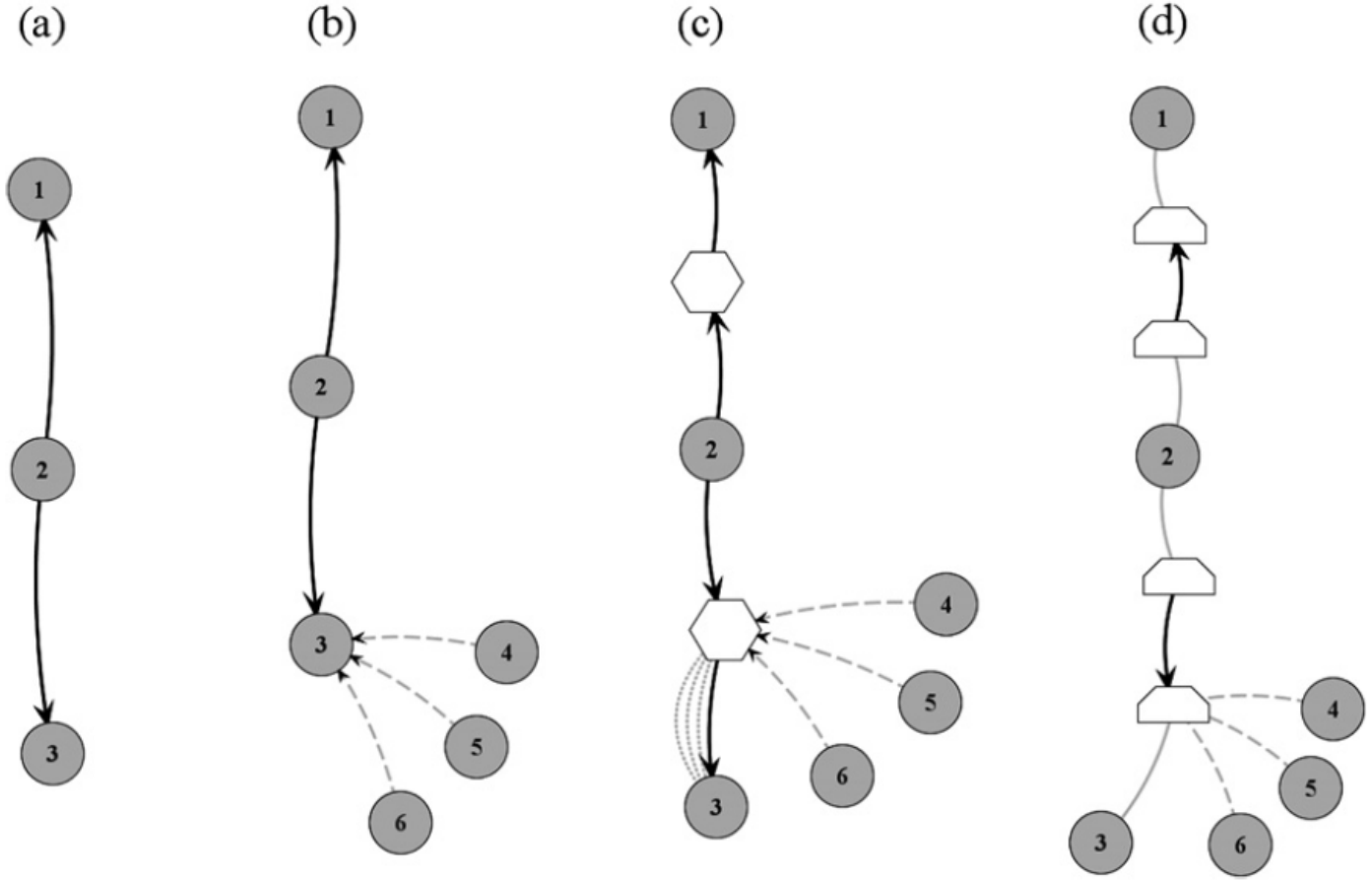
The system assigns scores to the entities of each connected component in the given network, by:

- Constructing different types of network representation;
- Detecting suspicious components based on structural and non structural characteristics;
- Detecting suspicious entities within the suspicious components, using the Iterative Assessment Algorithm (IAA);
- Visualization of results.

MODULE 1: REPRESENTATION

GUIDELINES

In order to choose the best network, you need to follow some guidelines, such as intention (appropriate for our aim), expressiveness, simplicity...



Four types of networks representing same two collisions – (a) drivers network, (b) participants network, (c) COPTA network and (d) vehicles network. Rounded vertices correspond to participants, hexagons correspond to collisions and irregular cornered vertices correspond to vehicles. Solid directed edges represent involvement in some collision, solid undirected edges represent drivers (only for the vehicles network) and dashed edges represent passengers. Guilt in the collision is formulated with edge's direction.

THE CHOICE

In the end, the best representations according to the authors were network b) and c): b) is going to be used for Module 2 and c) for Module 3.

Network a) is too simple while network d) involves unnecessary elements.

For the next steps, the connected components are simplified to detect communities -> based on edge betweenness.

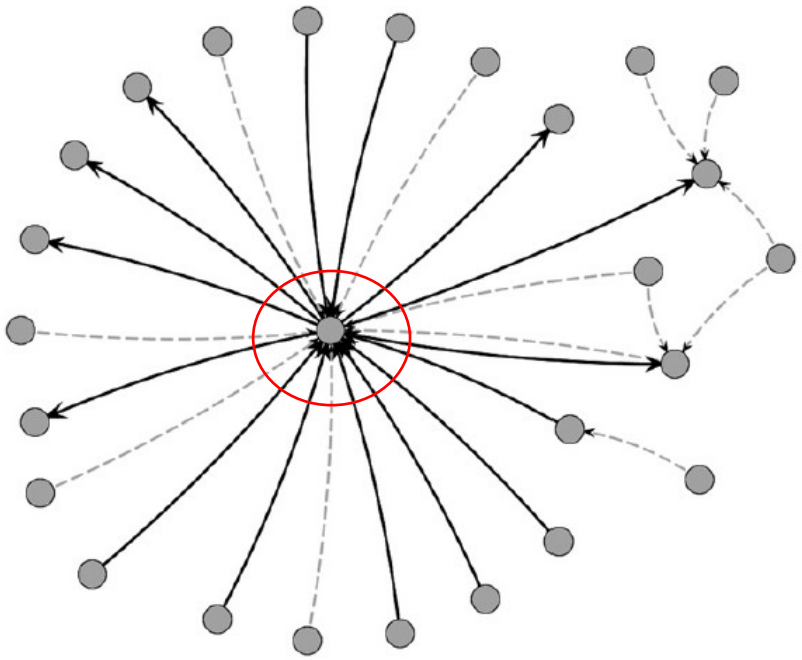
MODULE 2: SUSPICIOUS COMPONENT DETECTION

SUSPICIOUS STRUCTURAL ELEMENTS

Analysis have highlighted that fraudulent components share some structural elements:

- Larger than others;
- Denser;
- Almost every driver is linked to one collision;
- Some nodes have extremely high degree and centrality;
- Small diameter.
- Cyclic.

Example of a component of the networks (participants network) that shares some structural elements with fraudulent groups



HIGH DEGREE CENTRALITY NODE

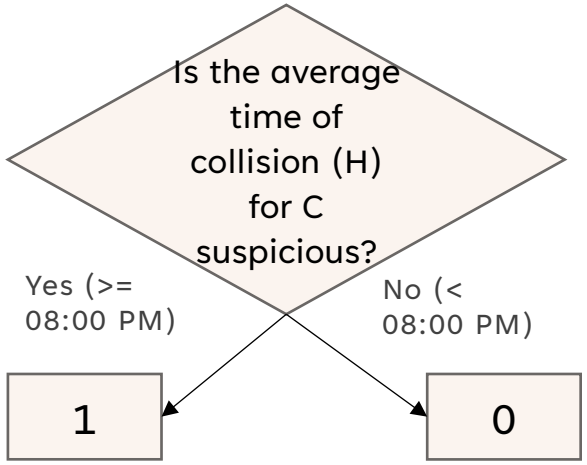
HIGH DENSITY

HOW TO ASSESS SUSPICION? INDICATORS

Each characteristic (H) of each component (C) is assigned a value for the indicator (I).

Two ways to deal with it:

1. For characteristics **independent** from the structure of the network, threshold is set by the expert;
2. For those depending on structure, a rewiring algorithm is employed.



Example of an indicator I **(non structural** based characteristic)

MODULE 2: SUSPICIOUS COMPONENT DETECTION

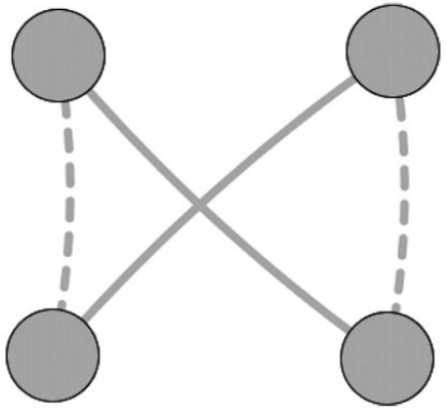
REWIRING ALGORITHM AND DEGREE MODIFICATION

This algorithm (iteratively) randomly chooses two edges in each component and switch one of their incident nodes.

Many repetition of the experiment.

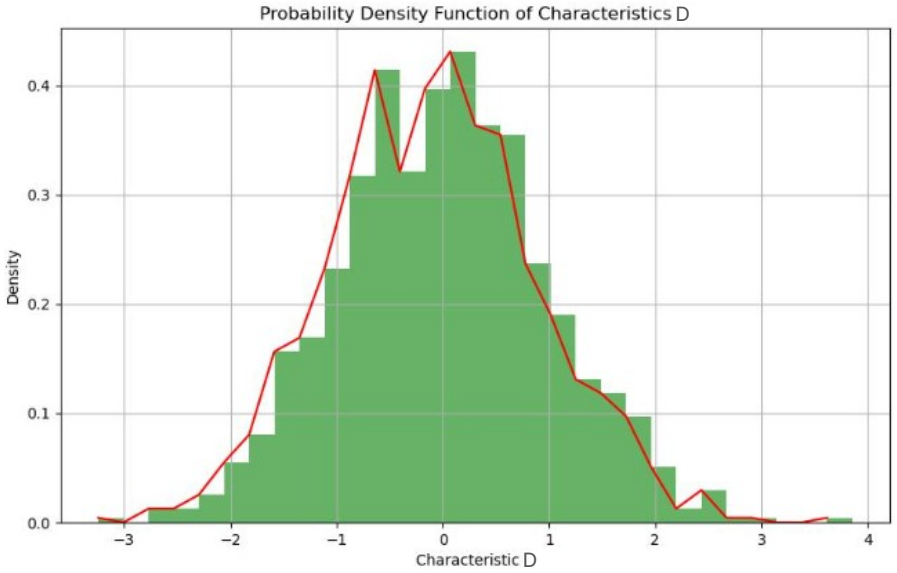
Before applying the rewiring, one extra node is created and linked with the original ones, in order to modify the final degree (otherwise it wouldn't).

200 random networks are built in this process.



Example of a rewired network in which dashed edges are replaced by solid edges

Example of the distribution for a generic characteristic D (elaborated with AI)



DEFINITION OF THE INDICATORS

The sample distribution for a generic characteristic (D) based on the obtained rewired networks is now calculated.

This distribution is considered "good"; values for non-suspicious D.

To assess if the indicator I gives 1 or 0, it is necessary to **test** if the original characteristic belongs to the sample distribution of "good" values (null hypothesis). If so, then it is a 0!

MODULE 2: SUSPICIOUS COMPONENT DETECTION

$$R_i(c) = \begin{cases} \hat{p}_i^0 & I_i(c) = 1, \\ -\hat{p}_i^1 & I_i(c) = 0, \end{cases}$$

Equation for RIDIT scoring function. P0 is the relative frequency of indicator I being 0 throughout all the components.

RIDIT FOR SCORING

After obtaining all the indicators, it's necessary to build a classifier.

it's also important to have a measure to «compare» the situation of the indicator in each component with the total indicators.

To do so, RIDIT method is employed. It is a scoring method that map categorical variables (the indicators in our case) in an interval [-1, +1].

At first, relative frequencies of indicators assuming values 1, 0 are calculated for the entire network. Then, the RIDIT score is computed for each indicator and each component: if component A has indicator I = 1 and the result for RIDIT is 20%, it means that having I = 1 is not significant (because it is already highly diffused).

MODULE 2: SUSPICIOUS COMPONENT DETECTION

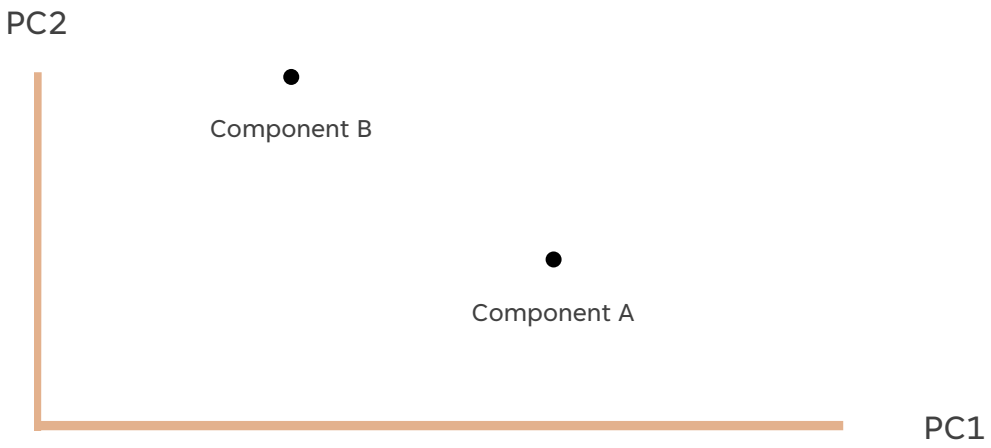
PRIDIT FOR WEIGHTING THE SCORES ⚖

Based on RIDIT, all the indicators would have the same weight!

To also cope with this problem, PRIDIT is used. PRIDIT combines RIDIT with Principal Component Analysis (PCA).

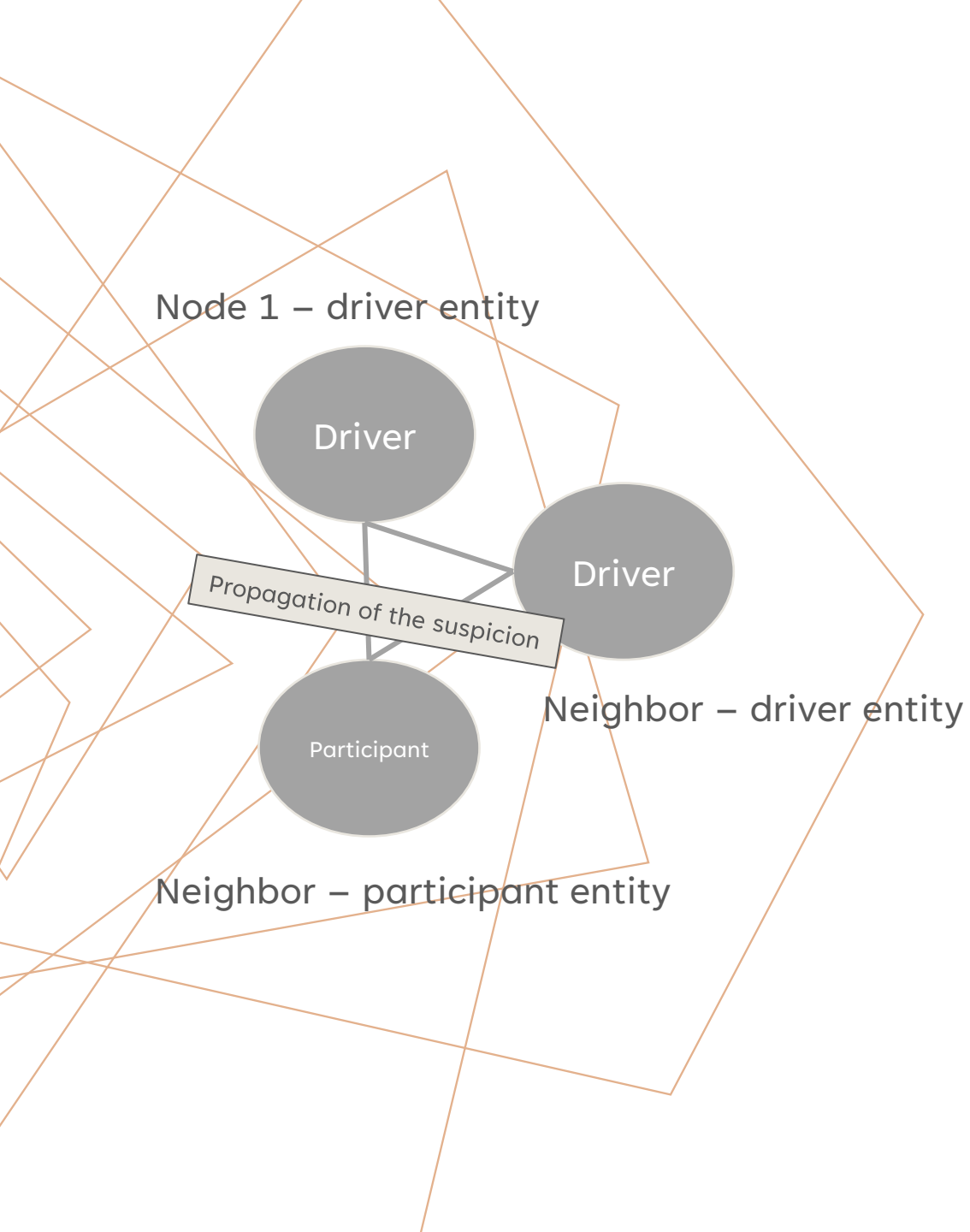
Starting from the matrix (on the right), PCA is employed and based on it the weights (that were randomly initialized) are updated. Greater weights are assigned to indicators which are more correlated to suspicious connected components.

Each component can now be scored, considering an established threshold (in this case, if >0 -> suspicious).



$$R = \begin{pmatrix} R_1(c_1) & R_2(c_1) & R_3(c_1) \\ R_1(c_2) & R_2(c_2) & R_3(c_2) \\ R_1(c_3) & R_2(c_3) & R_3(c_3) \end{pmatrix}$$

R = matrix where rows correspond to components (ci) and columns to indicators (i). r_i,j represent the value of RIDIT for each combination.



MODULE 3: SUSPICIOUS ENTITY DETECTION

ITERATIVE ASSESSEMENT ALGORITHM (IAA)

IAA is an algorithm that defines a scoring function for each entity in each component, that is generally formulated in this way: the starting function s(0) depends on some intrinsic and relational (related to neighbors) attributes. The suspicion score for the neighbors is modelled based on an **assessment model**.

The algorithm then iterates the the function based on neighbor scores, using a parameter α to weight the previous model relevance.

The optimal number of iterations k is dynamic based on the component (diameter of C).

Considering that in this module the COPTA network is used, the nodes related to collisions are considered as «bucket» as are not given a real score. «Non-bucket» nodes instead, have a true score that is going to be normalized.

MODULE 3: SUSPICIOUS ENTITY DETECTION

Three different assessment model are employed in this work.

AM_{raw}

RELATIONAL ATTRIBUTES ONLY

Sum of all suspicion scores for related entities.

AM_{bas}

RELATIONAL AND INTRINSIC

It adds to the RAW model the **factors.**

A factor is the multiplication of all *virtual factors for characteristics* belonging to an entity (choosen by the expert) and mapped in an interval

[-1, +1].

AM_{mean}

NORMALIZED RELATIONAL AND INSTRINSIC

The scores for neighbors are averaged, and the model (RAW or BAS) is multiplied for the average degree of the component.

DATA AND RESULTS

Dataset

Semi-structured data from police records; merge of one small labeled dataset and one larger unlabeled dataset.

3451 participants involved in 1561 collisions in Slovenia in years 1999-2008. Isolated cases were deleted.

Table 2
Class distribution for the data set used in the analysis of the proposed expert system.

	Count	*Proportion*	
Class distribution			
Fraudster	46	1.3%	21.8%
Non-fraudster	165	4.8%	78.2%
Unlabeled	3240	93.9%	

Results

The entire assessement is repeated 20 times and the scores were averaged.

At first, classification metrics are computed for the Golden standard (the model that combines PRIDIT and IAA_mean_bas).

Table 3
Performance of the system that uses *PRIDIT* analysis with IAA_{bas}^{mean} algorithm. Various metrics are reported; all except $\overline{AUC}$ are computed so the total cost (on the first run) is minimal.

Golden standard	
CA	0.8720
Recall	0.8913
Precision	0.6508
Specificity	0.8667
F1 score	0.7523
$\overline{AUC}$	**0.9228**

Table 4
Confusion matrix for the system that uses *PRIDIT* analysis with IAA_{bas}^{mean} algorithm (determined so the total cost on the first run is minimal).

	Suspicious	*Unsuspicious*
Confusion matrix		
Fraudster	41	5
Non-fraudster	22	143

DATA AND RESULTS

ASSESSMENT MODELS COMPARISON

The different models used in Module 3 are compared based on AUC.

Table 5
Comparison of different assessment models for *IAA* algorithm (after *PRIDIT* analysis).

Assessment models			
IAA_{raw}	IAA_{bas}	IAA_{raw}^{mean}	IAA_{bas}^{mean}
PRIDIT			
0.8872	0.9145	0.8942	0.9228

Table 7
Comparison of the *IAA* algorithm against several well known measures for anomaly detection in networks (on all the components and after *PRIDIT* analysis). For a fair comparison, no intrinsic attributes are used in the *IAA* algorithm (i.e. model AM_{raw}^{mean}).

IAA algorithm				
BetCen	*CloCen*	*DegCen*	*EigCen*	IAA_{raw}^{mean}
ALL				
0.6401	0.8138	0.7428	0.7300	0.8188
PRIDIT				
0.6541	0.8158	0.8597	0.8581	0.8942

ANOMALY MEASURES

Also in this case, AUC is used in order to compare the measures.

OTHER APPROACHES

Also in this case, AUC is used in order to compare the models.

Table 8
Comparison of different approaches for detection of fraudulent components (prior to no fraudulent entities detection and IAA_{bas}^{mean}).

Fraudulent components					
I_{MinCov}	$I_{I^{-1}}$	I_{BetCen}	*MAJOR*	*RIDIT*	*PRIDIT*
ALL					
0.6019	0.6386	0.6774	0.7946	0.6843	0.7114
IAA_{bas}^{mean}					
0.6119	0.8494	0.8549	0.8507	0.9221	0.9228

CONCLUSIONS

The Golden standard model seems to be the best in terms of performance, having a True Positive Rate of 90% and a True Negative Rate of 85% (labeled). For unlabeled data, the majority of the entities was considered non suspicious (many were isolated cases).

The system is able to detect fraudsters efficiently, based on intrinsic and relational attributes and focusing on larger suspicious group, also thanks to the way of representing data.

Also applicable in other domains; this system has to be used together with the domain expert's knowledge.

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THANKS FOR YOUR ATTENTION

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Data Analytics for Business and Society